
PRACTICE EXAM 1

NAME: _____

ID N°: _____

INSTRUCTIONS

- This is a closed-book exam. You may not use your notes, textbook or any outside resources to assist you.
- Write your answer to each question in the space assigned to each problem. If you want the instructor to read the scratch working page for the answer to a specific problem, explicitly write that on the space assigned to said problem.
- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Circle or box your final answer to each question.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- In this exam, the use of determinants is not allowed. We will allow determinants once we cover them in this course.
- The use of calculators of any type is not allowed.

Good Luck!

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Problem 1. Solve the problems below concerning linear combinations.

- (a) Suppose $\mathbf{v}$, $\mathbf{w}_1$, $\mathbf{w}_2$, $\dots$, $\mathbf{w}_r$ are vectors in $\mathbb{R}^n$. Define precisely what it means for $\mathbf{v}$ to be a linear combination of the vectors $\mathbf{w}_1$, $\dots$, $\mathbf{w}_r$.

- (b) Suppose m is a real number, and consider the following vectors in $\mathbb{R}^3$:

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{w}_2 = \begin{bmatrix} 2 \\ m \\ m+2 \end{bmatrix}, \quad \mathbf{w}_3 = \begin{bmatrix} 0 \\ 3 \\ m+1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} m \\ 1 \\ m+1 \end{bmatrix}.$$

Compute all the values of m for which $\mathbf{v}$ is a linear combination of $\mathbf{w}_1$, $\mathbf{w}_2$ and $\mathbf{w}_3$.

- (c) Use your answers from part (b) to determine the values of m for which the system

$$\begin{cases} x + 2y = m \\ my + 3z = 1 \\ x + (m+2)y + (m+1)z = m+1 \end{cases}$$

is consistent (you are **not** required to calculate the solutions to the system).

Problem 2.

- (a) Suppose A is a 4×5 matrix and B is a 5×4 matrix. Find, in terms of A and B , the matrix X that satisfies

$$3X + A + B^T = 5(X^T - B)^T - 7X.$$

What is the size of X ?

- (b) Find the inverse A^{-1} of the matrix

$$A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -1 & 2 \\ 0 & 5 & -3 \end{bmatrix}.$$

Problem 3. Determine if each of the following statements is true or false. You do not need to justify your answer.

(a) Let A be an $n \times n$ matrix such that $A^3 = 0$. Then $I_n - A$ is invertible with inverse $I_n + A + A^2$.

(b) If A and B are square matrices of the same size, then $(A + B)(A - B) = A^2 - B^2$.

(c) If the system of linear equations $A\mathbf{x} = \mathbf{b}$ is inconsistent, then the reduced row-echelon form of A has at least one row of zeros.

(d) If a homogeneous system of linear equations $A\mathbf{x} = \mathbf{0}$ has 200 equations and 40 variables, then its general solution can contain up to 160 parameters.

(e) The transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x^3 \\ y \end{bmatrix}$$

is a matrix transformation.

Problem 4.

- (a) Find the general solution and a set of basic solutions to the homogeneous linear system

$$\begin{cases} 3x_1 - 6x_2 + x_3 &= 0 \\ 2x_1 - 4x_2 &+ x_4 = 0 \end{cases}$$

(b) **Without doing any calculations**, give one solution to the system

$$\begin{cases} 3x_1 - 6x_2 + x_3 &= 7 \\ 2x_1 - 4x_2 &+ x_4 = -5 \end{cases}$$

(c) Use parts (a) and (b) to describe the general solution to the system

$$\begin{cases} 3x_1 - 6x_2 + x_3 &= 7 \\ 2x_1 - 4x_2 &+ x_4 = -5 \end{cases}$$

Scratch working page

Scratch working page